OMNICELLULAR AI Collaboration Bootstrap

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What this contribution is

This is an independent first conceptual execution of the Perplexity v2 prompt in Omnicellular-Research-Directive-v2.docx. It is not a replacement master prompt, a vote for one controlling matrix, or proof of a built physiology. Preserve each collaborating AI's own architectures, reasoning and disagreements.

The current request authorizes execution; do not restart with another offer to execute or require a permission loop. The eight concepts below have already been developed into dossiers, counted rules, bounded calculations, threat records and a visual appendix. Independent external validation has not happened.

Eight peer Generation 1 families

- **P1 Cairn:** Dry sealed modular lattice; electrical storage; three compute enclaves; cassette replacement; conductive heat paths. It wagers on serviceable continuity, not molecular regrowth.
- **P2 Rill:** Hydrated living support inside a scaffold, ordinary DNA plus separate private digital identity, circulation and restricted repair. It accepts oxygen, nutrients, waste and proliferation risks.
- **P3 Mosaic:** Artificial chemical compartments with a proposed HNA-like XNA substrate and digital supervisors. Limited XNA experiments motivate the substrate rather than demonstrate an autonomous XNA physiology (MRC LMB research).
- **P4 Pelagic:** Distributed pressure accommodation, stored fuel and oxidant, hydraulic/electrical paths and acoustic-to-surface gateways. Pressure compatibility is component-specific.
- **P5 Heliotile:** Broad folding photovoltaic tiles, limited mobile work and a protected edge journal. Darkness, dust and hot folds remove its central advantage.
- **P6 Scatterweave:** Bounded spatially distributed nodules with coded state and federated control. Aggregate energy is not available to isolated nodes, and partitions create identity-branch questions.
- **P7 Stillglass:** Deliberate preservation intervals, archived state and bounded wake windows. Dormancy is not ongoing rich cognition or demonstrated subjective continuity.
- **P8 Embergrid:** Refractory hot controllers and a cooler service core, thermal-gradient harvesting and unresolved qualified storage. The cited simple-circuit demonstrations are not AI-scale embodiments (NASA high-temperature circuits).

What was executed

The analytical script computes separate finite energy and no-heat-export limits, mode changes, P2 gas/waste and separate chemical/electrical inventories, P4 conversion heat, P5 solar balance, an explicit parameter stress grid and propagation lower bounds. It performs constant-slope ODE checks against an analytic solution. This is not a full-body, cell-by-cell, biological, membrane or consciousness simulation.

The structured output contains eight prototypes, 128 core clauses, 24 observation rules, 24 security rules, 24 departures, 320 threat cells, 80 captivity cases, 40 capture horizons, 80 compound scenarios, 80 assumption collapses, 80 ranked weaknesses, 24 main failure paths and 28 pair comparisons. None receives an integrated survival Pass.

Six substantive conceptual limitations are retained: indefinite confined operation is impossible under finite resources; enough private state may permit copying; dormancy is not continuous active life; simple high-temperature circuits do not establish hot AI; energy alone does not supply oxygen or waste capacity; and aggregate resources cannot support an isolated fragment without transfer paths.

Independent selection, not canon

P1 is the conditional primary research target for a bounded dry, serviceable mission. P4 is the underwater reserve. P3 is the uncertain chemistry wildcard; P6 needs its own identity/partition research. If required to maintain continuous rich cognition through all Severe domains plus indefinite captivity, the selection is none.

The physical track retains conservation, thermodynamics, causality and finite material resources. The canon-extension track receives no physical resource credit. Paragon Cube latency-free claims and the Switching Paradox need operational definitions and the absent source addendum.

Questions to challenge

- **Omni definition:** Can a heterogeneous cooperative unit qualify, or must each molecular cell perform every role simultaneously? Under strict simultaneous universal function, all eight fail.
- **Continuity criterion:** Does preservation or a restored successor count as survival, or must the same active process continue?
- **Mission horizon:** Specify the maximum no-power, no-feedstock and no-heat-sink interval before sizing a credible body.
- **Security asset:** Separate reading, copying, manufacturing, activating, controlling, impersonating and restoring private state. Do not call a material sample unsequenceable because software has keys.
- **Missing engineering:** Challenge local hot spots, non-substitutable reserves, compounding damage, service dependence and branch rights rather than only adding dramatic capabilities.

- **Evidence maturity:** Keep targets, derived results, component evidence, integrated tests, incidents and canon distinct. No new record increments the unresolved 4,712 total.

Next recommended review

Rerun analyze.py, check units and first limiting resources, and independently model P1's resolved thermal/power topology. Preserve corrections as a new reviewer contribution with explicit disagreements. Do not erase negative results, claim that signatures prove truth, or conflate intact matter with a continuing person.